

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Duration: 60 Minutes

Total Marks: 20

Code of Conduct

I MOHAN RAM / 192521117 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Mohan Ram

Criteria	Concept Identification (1.25)	Relationship Mapping (1.25)	Hierarchy and Organization (1)	Technical Relevance (1)	Visual Clarity (0.5)	TOTAL (5)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Architecture: Include front-end, back-end, client devices, data centers, and distributed servers.
2. Concept Map 2: Characteristics of Cloud Computing: Connect on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
3. Concept Map 3: Deployment Models: Map public cloud, private cloud, hybrid cloud, and community cloud with usage scenarios.
4. Concept Map 4: Cloud Benefits vs Challenges: Show advantages (cost efficiency, scalability, flexibility) and limitations (security, downtime, vendor lock-in).

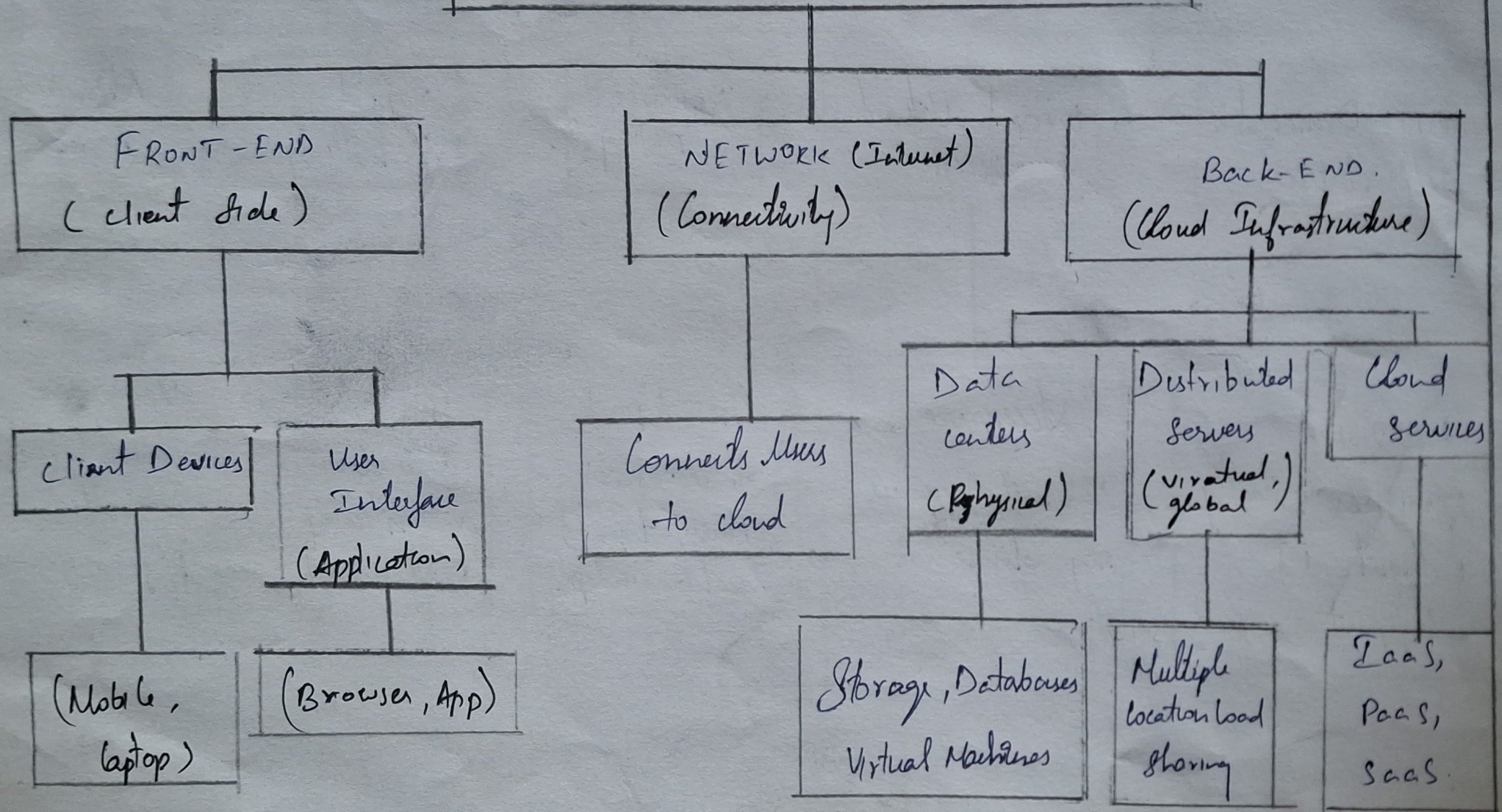
Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

CLOUD COMPUTING ARCHITECTURE



CHARACTERISTICS OF CLOUD COMPUTING

ON-Demand
Self-service

Provision
resources instantly
without IT
support

Broad Network
Access

Access via
Internet Anywhere
any device

Resource
pooling

Shared pool
of resources
(Multi-tenant)
location indep.

Rapid
elasticity

Scale up &
Scale down
automatically
based on need

Measured
Service

Usage is
monitored
& billed
(pay-as-you-use)

CLOUD DEPLOYMENT MODELS

Public cloud

Private cloud

Hybrid cloud

Community cloud.

Owned by provider (3rd party)

Owned by single org (internal)

Combination of public & private clouds

Shared by multiple orgs (common goals)

Used by general public over internet

Usage: Banking, healthcare

Usage: E-commerce, seasonal load.

Usage: Govt sector's, research groups

Example: Web hosting, apps, storage.

Used by one organization for security.

Data & apps move b/w clouds

Organizations with similar requirement.

CLOUD COMPUTING : BENEFITS VS CHALLENGES

Advantages

Cost
Efficient

Reduced
IT cost
Pay-as-
you-use

Scalability

Scale up/
down based
on demand

Flexibility

Access anytime
anywhere,
any device

Challenges

Security

Data risks
& privacy
concerns

Downtime
issues

Service
outages
(failures)

Vendor
lock-in

Dependency on
single provider
difficult to switch.